ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-01 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

Rules A and B are $2x + 1$ and $x + 3$. At which input do they give the same output?

2. I can use input-output rules to represent relationships.

Complete the table.

x	-2	0	4
$y = 3x - 1$	?	?	?

3. I can use input-output rules to represent relationships.

Write a rule.

Inputs 0,1,2 give outputs 4,7,10.

4. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Is $(1, 3), (2, 3), (3, 3)$ a function? Evaluate $2(-3) + 5$.

6. I can distinguish functions from relationships that are not functions.

The pairs are $(1, 4), (2, 6), (1, 7)$. Is this relation a function? Explain.

7. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

8. I can distinguish functions from relationships that are not functions.

Identify the repeated input.

x	-1	2	-1
y	3	5	8

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

For $f(x) = x + 2$ with inputs $\{-1, 0, 3\}$, list its outputs. Then extend 4, 9, 14, … one term.

10. I can identify the domain and range of a relation.

List the range.

$(2, 5), (4, 7), (6, 5)$

11. I can identify the domain and range of a relation.

Tickets cost $6 each, and at most 8 tickets may be bought. What inputs are reasonable for the total-cost rule?

12. I can identify the domain and range of a relation.

List the domain.

$(2, 5), (4, 7), (6, 5)$

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

Write an equation.

Slope 4; y-intercept −2.

14. I can connect linear growth in tables, graphs, and equations.

x	0	1	2	3
A	1	4	7	10
B	1	4	9	16

Which is linear? Give numerical evidence.

15. I can connect linear growth in tables, graphs, and equations.

Find the change per step for outputs 5, 9, 13, 17. Is $(1, 2), (2, 2), (1, 3)$ a function?

16. I can connect linear growth in tables, graphs, and equations.

Find the slope.

Points (1,3) and (5,11)

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

Find the run.

$(-1,8)$ to $(6,10)$

18. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0, 2)$ and $(3, 8)$.

19. I can calculate and interpret slope as a constant rate of change.

As x rises by 4, y falls by 12. Find the slope.

20. I can calculate and interpret slope as a constant rate of change.

Find the slope.

$(-2,1)$, $(2,9)$